

Project

(Due on: May 13, 2026 at mid-night)
(Submit on Canvas as one .zip file)

In this project, each team needs to pick one of the following ideas to work on.

Project 1: Assessing Neurological States from Physiological Signals

Neurological states including different types of stress (physical, emotional and cognitive) could be recognized using different types of physiological signals. In this project, the aim is to identify neurological states using non-brain physiological signals such as electrodermal activity, accelerometer data, heart rate and many others. This could simplify the process of detecting neurological states.

Dataset Source: <https://physionet.org/content/noneeg/1.0.0/>

Project 2: Sleep Stage Prediction from Heart Rate and Motion Data

Insomnia represents one of the most irritating challenges people might have given that humans ability to function depends primarily on the quality of their sleep. In this project, data mining methods will be developed to recognize different sleep stages using heart rate and motion data recorded using wearable sensors.

Dataset Source: <https://physionet.org/content/sleep-accel/1.0.0/>

Project 3: Classification of Neurodegenerative Diseases using Gait Analysis

Diagnosis of different neurodegenerative diseases represents one of the most challenging tasks in medicine. While brain scans and blood tests have been shown to help in such problem, other methods that involve analysis of moving patterns of patients have been proposed. In this project, data mining methods will be developed to analyze and classify gait information of three different neurodegenerative diseases; namely Parkinson's, Huntington's and Amyotrophic Lateral Sclerosis.

Dataset Source: <https://physionet.org/content/gaitnidd/1.0.0/>

Project 4: Predicting Intensive Care Unit (ICU) Mortality

This project aims at predicting which patients of the ICU are at high risk of mortality versus patients who could be cured at the ICU. This can help hospitals manage resources in a better way to help patients at high risk.

Dataset Source: <https://physionet.org/content/challenge-2012/1.0.0/>

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Project 5: Predicting Football Games Result using In-game Events

This project aims at predicting the results of the football games based on in-game events. Different feature extraction and classification techniques will be examined in order to predict during the game how the game will end. This can help coaches change their in-game tactics to avoid losing a game.

Dataset Source: https://figshare.com/collections/Soccer_match_event_dataset/4415000/3

Project 6: Bank Marketing Data

The dataset represents direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to assess if the product (bank term deposit) would be ('yes') or not ('no') subscribed.

Dataset Source (Requires Login): <https://data.world/data-society/bank-marketing-data>

Project 7: US Accidents Dataset

This dataset contains detailed information about traffic accidents across the United States, collected from multiple real-time traffic APIs. It includes features such as location, weather conditions, visibility, road characteristics, and severity levels, making it highly valuable for transportation analysis and safety modeling. The dataset can be used for accident severity prediction, temporal trend analysis, and understanding the impact of environmental factors on road incidents.

Dataset Source: <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

Project 8: Hotel Booking Demand Dataset

This dataset includes booking information for a city hotel and a resort hotel, capturing details such as reservation status, lead time, customer demographics, pricing, and stay duration. It reflects real-world hotel booking behavior and demand patterns, making it suitable for tasks like cancellation prediction, revenue optimization, customer segmentation, and demand forecasting. The dataset provides insights into seasonal trends, booking channels, and guest preferences in the hospitality industry.

Dataset source: <https://www.kaggle.com/datasets/jessemotipak/hotel-booking-demand>

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Deliverables:

- Your Python code (.py or Jupyter notebook) to be submitted on Canvas on **May 13 at mid-night**
- A report (to be submitted on Canvas on **May 13 at mid-night**) that includes the following:
 - Introduction: A description of the problem the selected project addresses
 - Methods: A description of the approach followed to analyze the dataset
 - Results: A description of the achieved results
 - Conclusions
- Submission of the above items should be done as one .zip file by the deadline
- A presentation (to be presented in class on **May 14**). The presentation should follow the same flow as the submitted report.

Guidelines:

- This is a group project. A maximum of 3 students per group is allowed.
- Each team must send an e-mail by **Wednesday, April 22 at mid-night** specifying the members of the team and the name of the selected project.
- **Changing teams or changing the dataset of the project will not be allowed.**
- Project grading will be as follows (out of 20):
 - 5 points on the code or project submitted
 - 5 points on the submitted report
 - 5 points on the presentation
 - 5 points on the depth of the analysis and the obtained insights